

EE5203 L05 Study Sheet

GPIO and Hardware Abstraction - Basri Erdogan

Essential question: HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, GPIO_PIN_SET) and GPIOA->BSRR = (1U << 5) produce the same hardware result — what exactly does the HAL layer add, and how do you verify it wrote the bits you expect?

What you should be able to do

- Read a GPIO_InitTypeDef configuration and predict the register fields it produces.
- Map GPIO_MODE_OUTPUT_PP, GPIO_NOPULL, GPIO_SPEED_FREQ_LOW to MODER/OTYPER/PUPDR/OSPEEDR bits.
- Use HAL_GPIO_WritePin, TogglePin, ReadPin correctly.
- Explain floating inputs, pull-up/pull-down, and why B1 is active-low.
- Detect a button **edge** (with debounce) rather than a level.
- Verify any HAL configuration in the SFR view.

The abstraction stack

```
your code -> STM32 HAL -> CMSIS register -> hardware pin
led_on()      HAL_GPIO_WritePin()  GPIOA->BSRR  PA5 / LD2
```

HAL changes how you *write* the command, not which hardware receives it. Neither interface removes the need to verify the result.

Configuration: GPIO_InitTypeDef

```
GPIO_InitTypeDef config = {0};          /* zero every member first      */

config.Pin    = GPIO_PIN_5;
config.Mode   = GPIO_MODE_OUTPUT_PP;
config.Pull   = GPIO_NOPULL;
config.Speed  = GPIO_SPEED_FREQ_LOW;

HAL_GPIO_Init(GPIOA, &config);          /* address: the function READS
                                         your members through it      */
```

HAL name	Register effect (PA5)
GPIO_PIN_5	selects pin 5 (a mask, like $1U \ll 5$)
GPIO_MODE_OUTPUT_PP	MODER[11:10] = 01, OTYPER[5] = 0
GPIO_NOPULL	PUPDR[11:10] = 00
GPIO_SPEED_FREQ_LOW	OSPEEDR[11:10] = 00

For PC13 the same fields live at bits 27:26 ($13 \times 2 = 26$).

Speed is not blink rate: Speed sets the electrical transition strength; HAL_Delay sets the visible timing.

Runtime: the three functions

Function	Job	Returns
HAL_GPIO_WritePin(port, pin, state)	force high/low	—
HAL_GPIO_TogglePin(port, pin)	reverse output	—
HAL_GPIO_ReadPin(port, pin)	sample input	GPIO_PIN_SET / GPIO_PIN_RESET

Configuration happens once (MX_GPIO_Init(), called by main() before the loop); these run inside the loop.

Inputs: floating, pulled, and B1

- **Floating:** nothing defines the level — noise decides what you read.

- **Pull-up** (PUPDR = 01): released reads 1. **Pull-down** (10): released reads 0. **None** (00): the external circuit must decide.
- **B1 on PC13** has an **external** pull-up (R30) and grounds the pin when pressed → configure `GPIO_NOPULL`, and the logic is **active-low**:

B1	PC13	HAL_GPIO_ReadPin
released	high	<code>GPIO_PIN_SET</code>
pressed	low	<code>GPIO_PIN_RESET</code>

Edge detection, not level

```
static uint32_t prev = 1U; /* released */
uint32_t cur = (uint32_t)HAL_GPIO_ReadPin(GPIOC, GPIO_PIN_13);

if (prev == 1U && cur == 0U) { /* falling edge = new press */
    ++press_count;
    HAL_Delay(10); /* debounce */
}
prev = cur;
```

Testing `cur == 0U` alone counts continuously while the button is held; the `prev/cur` pair fires once per press. The short delay rides out contact bounce.

Ownership rule

CubeMX owns generated init; USER CODE regions own your calls; the SFR view provides the evidence; direct register writes only when the exercise assigns them. Never let CubeMX and hand-written register code fight over one pin.

Common mistakes

- `MX_GPIO_Init()` never runs (or is called twice) before the loop.
- Port/pin mismatch: `GPIOA` with `GPIO_PIN_13`, or `GPIOC` with `PIN_5`.
- Wrong pull for the circuit — internal pull-up added on `PC13` is harmless but wrong-by-specification; missing pull on a truly floating input is a bug.
- Reversed logic: treating `GPIO_PIN_SET` as pressed.
- Edits outside USER CODE regions vanish on regeneration.
- Confusing **Speed** with blink timing.

Mastery self-check

- ☐ I can write the four `GPIO_InitTypeDef` members for `PA5` output from memory.
- ☐ I can give the `MODER/PUPDR` field values HAL writes for `PC13` input, no pull.
- ☐ I can explain why `HAL_GPIO_Init` takes `&config`.
- ☐ I can state what `B1` pressed reads and why.
- ☐ I can write the edge-detection pair and explain the debounce delay.
- ☐ I can list the three SFR checks that verify the lab's configuration.
- ☐ I can argue one strength and one cost of each interface.

Five review questions

1. `HAL_GPIO_Init(GPIOC, &config)` ran with `Mode = GPIO_MODE_INPUT`, `Pull = GPIO_NOPULL`, `Pin = GPIO_PIN_13`. Which register fields changed, and to what values?
2. Why does passing `config` (without `&`) not compile — and if it did, what Week-4 concept explains why it still could not work?
3. A student configures `PC13` with `GPIO_PULLUP`. Does the button still work, and what physically changes on the pin?
4. The press counter increases by 3 for one press. Name the two distinct causes and the fix for each.
5. In `stm32f4xx_hal_gpio.c`, `HAL_GPIO_WritePin` ends in a single `BSRR` write. Why does that make `plain = safe` there — and what does this tell you about what HAL is *made of*?

See the L05 worksheet for extended practice. Worked solutions are released after the practice window.